

$$\therefore \phi_B = -470 \text{ J kg}^{-1} \text{ and } \phi_A = -495 \text{ J kg}^{-1}$$

13 A Using $Q = mc\Delta\theta$,

$$\text{Rate of heat removed is } \frac{Q}{t} = \frac{m}{t} c \Delta\theta$$

$$\Rightarrow \frac{m}{t} = \frac{\left(\frac{Q}{t}\right)}{c\Delta\theta} = \frac{\left(\frac{4.0 \times 10^{11}}{60 \times 60}\right)}{4200 \times 8} = \frac{4.0 \times 10^{11}}{4200 \times 8 \times 60 \times 60} \text{ kg s}^{-1}$$

14 C For the grain of sand